AP CALCULUS AB - UNIT 1

1.4 Estimating Limit Values from Tables

Lesson Overview

- Select input values that approach the target from both sides.
- Estimate finite, infinite, and nonexistent limits from tabular patterns.
- Distinguish the center value from nearby evidence.
- Evaluate the reliability of a table, including rounding, spacing, and deceptive sampling.

Key Knowledge and Vocabulary

Direction. Inputs less than the target estimate the left-hand limit; inputs greater than the target estimate the right-hand limit.

Closeness. Use several inputs increasingly close to the target, not one isolated row.

Stable digits. Report only digits that remain stable as inputs move closer.

Center value. A row at $x = c$ records $f(c)$, not the limit.

Evidence, not proof. A finite table supports an estimate but cannot by itself prove all nearby behavior.

Explanation and Strategy

Step 1. Build two columns of inputs: $c - 10^{-k}$ and $c + 10^{-k}$.
Step 2. Compare both sides separately before forming a two-sided conclusion.
Step 3. Use calculator precision high enough to avoid rounding a hole, jump, or oscillation into a false pattern.

Solved Examples

Worked Example: Finite estimate

Estimate the limit from the table.

x	3.9	3.99	3.999	4.001	4.01	4.1
$f(x)$	7.81	7.9801	7.998001	8.002001	8.0201	8.21

Solution. Both sides approach 8. The stable digits support $\lim_{x\to 4} f(x) = 8$.

Worked Example: One-sided disagreement

Analyze the table.

x	-2.01	-2.001	-1.999	-1.99
$g(x)$	4.98	4.998	-0.997	-0.970

Solution. From the left, values approach 5; from the right, they approach -1. Therefore the two-sided limit does not exist.

Worked Example: Hidden center value

A table includes $f(2) = 100$, while rows at $1.99, 1.999, 2.001, 2.01$ are near 3. What should be concluded?

Solution. The nearby evidence supports $\lim_{x\to 2} f(x) = 3$, while $f(2) = 100$. The center row does not determine the limit.

Leveled Practice

Shared Representation / Data

x	3.9	3.99	3.999	4.001	4.01	4.1
$f(x)$	7.81	7.9801	7.998001	8.002001	8.0201	8.21

x	−2.01	−2.001	−1.999	−1.99
$g(x)$	4.98	4.998	−0.997	−0.970

Easy Level

Foundational fluency. Focus on notation, definitions, and direct procedures.

1. Use the first shared table to estimate $\lim_{x\to 4^-} f(x)$.

2. Use the first shared table to estimate $\lim_{x\to 4^+} f(x)$.

3. Use the second shared table to estimate $\lim_{x\to -2^-} g(x)$.

4. Use the second shared table to estimate $\lim_{x\to -2^+} g(x)$.

Medium Level

Apply the idea in one or two connected steps.

5. Create four suitable x-values to estimate a two-sided limit at $x = 7$.

6. A table uses only $x = 1.1, 1.01, 1.001$. Which limit direction can it directly estimate at $x = 1$?

7. Values are $2.4, 2.49, 2.499$ from the left and $2.6, 2.51, 2.501$ from the right. Estimate the two-sided limit.

8. Why is the row $x = c$ usually omitted when constructing a limit table?

Hard Level

Connect representations, conditions, and justification.

9. A calculator rounds all outputs to two decimals and reports 1.00 on both sides. Explain why this does not prove the limit is exactly 1.

10. Near $x = 0$, a table alternates $0.8, -0.9, 0.95, -0.98$ as inputs get closer. What behavior is suggested?

11. Near $x = 3$, magnitudes grow 10, 100, 1000 on both sides, all positive. Write the suggested limit.

12. Near $x = 3$, the left values are $-10, -100, -1000$ and right values are $10, 100, 1000$. State the conclusion.

Difficult Level

Use nonroutine structure and precise mathematical reasoning.

13. A table samples $f(x) = \sin(\pi/(x-1))$ only at $x = 1 + 1/n$, where every output is 0. Why is the table deceptive?

14. Design a two-sided table that would distinguish $|x|$ from x near 0.

15. A table for $\frac{1-\cos x}{x^2}$ near 0 stabilizes at 0.500000 until cancellation error appears at extremely small x. Explain the numerical issue.

Challenge Level

Synthesize, construct, generalize, or prove.

16. Give two functions that agree at every input in the finite table $x = \pm 0.1, \pm 0.01, \pm 0.001$ but have different limits at 0.

17. Explain how two interlaced sequences can disprove a proposed table-based limit.

AP-Style Practice

Multiple-Choice Practice – No Calculator

Choose the best answer. Use exact values unless a decimal approximation is requested.

1. A table near $x = 5$ has outputs approaching -2 from both sides, while the center row is 17. Which conclusion is best?

(A) $\lim_{x\to 5} f(x) = 17$
(B) $\lim_{x\to 5} f(x) = -2$
(C) The limit DNE.
(D) No conclusion is possible.

2. Which input list is best for estimating $\lim_{x\to 3} f(x)$?

(A) $0, 1, 2, 3$
(B) $2.9, 2.99, 3.01, 3.1$
(C) $3, 4, 5, 6$
(D) $-3, -2, -1, 0$

3. A finite table alone can

(A) prove a limit for every real input near the target
(B) support an estimate of a limit
(C) determine the function value when it is omitted
(D) guarantee continuity

Multiple-Choice Practice – Calculator Required

Choose the best answer. Use exact values unless a decimal approximation is requested.

4. A calculator table for $h(x) = \frac{e^x - 1}{x}$ gives 0.99950, 0.99995, 1.00005, 1.00050 near $x = 0$. The best estimate is

(A) 0 **(B)** 0.5
(C) 1 **(D)** DNE

5. For $p(x) = \frac{1 - \cos x}{x^2}$, a calculator reports 0 at $x = 10^{-10}$ after reporting values near 0.5 for larger inputs. The most likely explanation is

(A) the exact limit changes at smaller scales **(B)** floating-point cancellation error
(C) a jump at 0 **(D)** the function becomes linear

Free-Response Practice – No Calculator

A table for a function F is shown.

x	1.9	1.99	1.999	2.001	2.01	2.1
$F(x)$	−0.71	−0.9701	−0.997001	−1.003001	−1.0301	−1.31

(a) Estimate $\lim_{x \to 2} F(x)$. [2 points]

(b) Does the table determine $F(2)$? Explain. [2 points]

(c) State one limitation of the numerical evidence. [2 points]

Free-Response Practice – Calculator Required

Use a calculator to investigate $G(x) = \dfrac{\sqrt{x+4} - 2}{x}$ near $x = 0$.

(a) Create a two-sided table with at least three inputs on each side and estimate the limit. [2 points]

(b) Explain why direct calculator entry at $x = 0$ does not answer the limit question. [2 points]

(c) Use algebra to confirm the estimate. [3 points]